

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Chandubhai S Patel Institute of Technology

3rd Semester B.Tech.(CE/IT)

University Exam April-2012

MA-201 Discrete Mathematics

Date: 30/04/'12, Monday

Time: 01:30 pm to 04:30 pm

Maximum Marks: 70

Instruction:

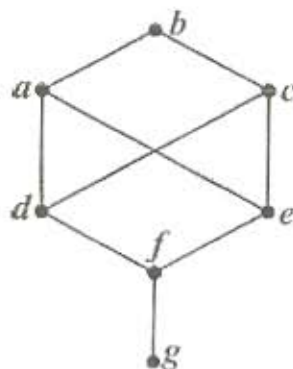
1. This question paper contains two sections Section-I and Section-II.
2. Section-I carries 35 and Section-II carries 35 marks.
3. Make suitable assumption whenever necessary and state them.
4. Figure to the right indicates the distribution of full marks for the respective question.

Section-I

Q-1 Attempt the following::

[07]

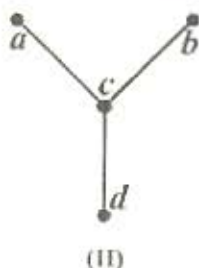
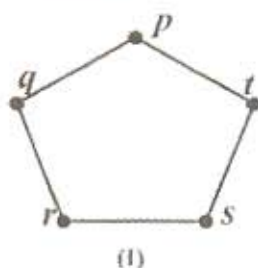
1. When four 1's are taken as a group on a Karnaugh map, the number of variables eliminated from the output expression is ____
2. Boolean algebra simplifies logic circuits. True or False?
3. In Boolean algebra, the word "literal" means ____
4. Symbolize the expression: There does not exist a woman who has taken a flight on every airline in the world.
5. Is there an infinite poset with a least element? Justify your answer.

Q-2 [(A)] Define POSET. Consider the Poset $(A, \leq)$ given by the following Hasse diagram. [05]

1. List all the minimal elements of $x (A, \leq)$
2. List all the upper bounds of the set $\{c, d\}$.
3. List a pair of incomparable elements of A .
4. Is $x (A, \leq)$ a lattice? Why or why not?

(B) Define Lattice. Which of the following Hasse diagrams represent lattices?

[05]



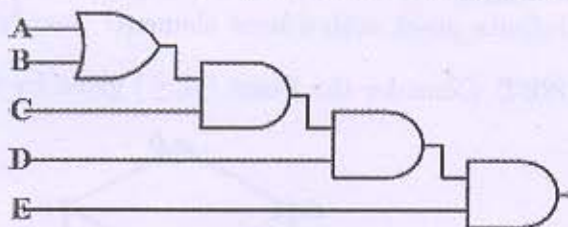
- (C) Is it true that every poset has a greatest element? Justify your answer. [02]
 (D) Let the universe be the set of airplanes and let $F(x, y)$ denote "x flies faster than y". Translate the following expression into English:
 (a) $\forall x \exists y F(x, y)$ (b) $\exists x \forall y F(x, y)$ [02]

OR

- Q-2(A) Let L be a lattice then for every 'a' and 'b' in L, $a \vee b = b$ if and only if $a \leq b$. [04]
 (B) Draw the Hasse diagram of the Poset $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ with the relation $a \leq b$ if and only if 'a' divides 'b'. [04]
 (C) Consider the poset $(\{1\}, \{2\}, \{4\}, \{1, 2\}, \{1, 4\}, \{2, 4\}, \{3, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, \leq)$
 (i) Find the maximal elements. (ii) Find the minimal elements. (iii) Is there a least element. (iv) Find the least upper bound of $\{\{2\}, \{4\}\}$, if it exists. [04]
 (D) Is it true that every finite lattice has a least element? Justify your answer. [02]
- Q-3(A) What is the Karnaugh map? Use Karnaugh map to simplify the Boolean expression:
 (i) $X'Y'Z' + XYZ + X'Y'Z + X'YZ$ (ii) $A'C + A'B + AB'C + BC$ [06]
 (B) Simplify the given Boolean expression: $AB + (AC)' + AB'C(AB + C)$ [04]
 (C) Using Boolean algebra show that $abc + abc' + ab'c + a'bc = ab + ac + bc$ [04]

OR

- Q-3(A) What is the Karnaugh map? Use Karnaugh map to simplify the Boolean expression:
 $WXY'Z' + WX'YZ + WX'YZ' + WX'Y'Z' + W'XY'Z' + W'X'YZ' + W'X'Y'Z'$ [06]
 (B) Derive the Boolean expression for the logic circuit shown below: [04]



- (C) Convert the following SOP expression to an equivalent POS expression:
 $ABC + AB'C' + AB'C + ABC' + A'B'C$ [02]
 (D) Determine the values of A, B, C and D that make the sum term $A' + B + C' + D$ [02]

Section-II

Q-4 Do as Directed: [07]

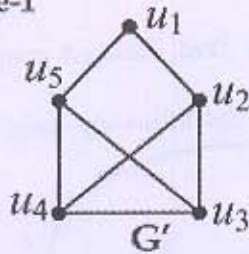
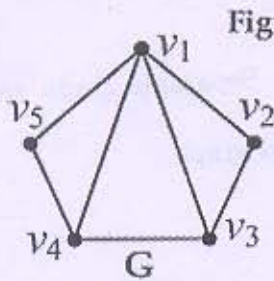
1. State Euler's theorem for planar graph.
2. The maximum number of edges in a simple graph with n vertices is ____
3. Define Connected graph with example.
4. Define Tree with example.
5. $K_{3,3}$ is a planar graph. True/False
6. The number of vertices in a binary tree is always odd. True/False.
7. In the adjacency matrix of a graph, the number of 1's in each column equals the degree of the corresponding vertex. True/False

Q-5 (a) Draw the separate graph for each [05]

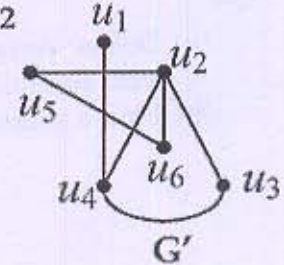
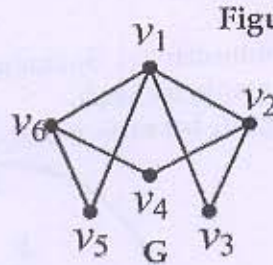
1. Hamiltonian circuit but not Eulerian circuit.
2. Neither Hamiltonian circuit nor Eulerian circuit.
3. Eulerian circuit but not Hamiltonian circuit.

(b) Define Adjacency matrix, Incidence Matrix with two examples. [04]

(c) Define Isomorphic graph. Are the following graph G and G' are isomorphic? [05]



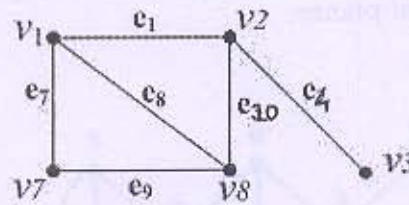
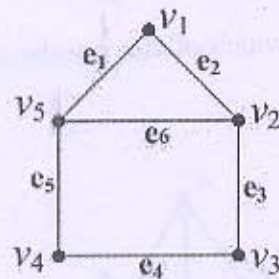
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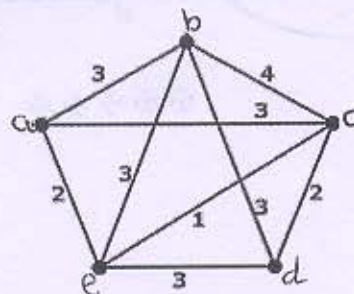
Q-5

(a) Define Complement of graph. Draw the complement of K_5 . [03]

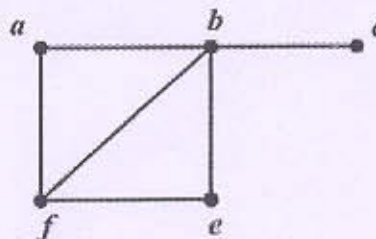
(b) Find Union, intersection and Ring sum of the given pair of graph. [04]



(c) Find the minimal Spanning tree by kruskal algorithm for the following weighted graph. [03]



(d) Find the number of Vertices, the number of Edges and the degree of each vertex in the given graph. [04]



Q-6

(a) Define Binary tree with example. Is it possible to draw a binary tree with 13 vertices which has a vertices of the degree 1 or 3. [03]

(b) Define Bipartite graph with example and draw $K_{1,2}$ and $K_{3,2}$. [03]

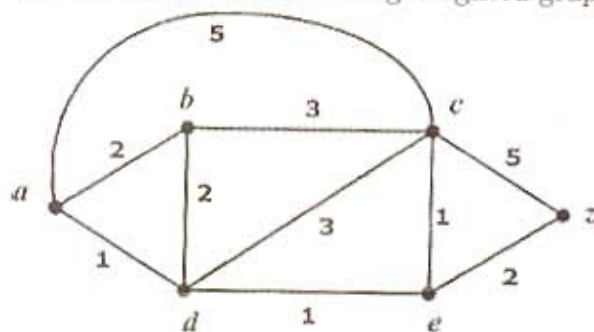
(c) Define path, subgraph, Loop, Multigraph with example. [04]

- (d) A Connected planar graph has seven vertices having degrees 3,3,2,3,3,2 and 4. How many Edges are there? How many Regions are there? Draw two such graph if exists. [04]

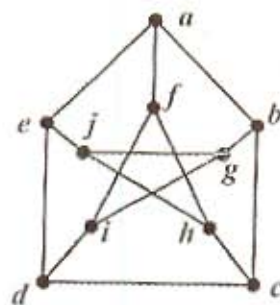
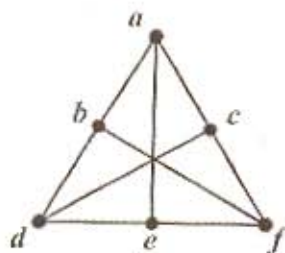
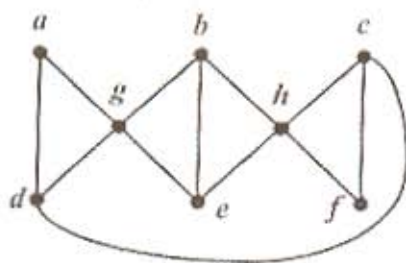
OR

Q-6

- (a) Define Vertex connectivity, Spanning Tree, Rooted graph, Separable graph and Block with suitable example for each. [05]
 (b) Find a shortest path between a and z the following weighted graph. [04]



- (c) State Kuratowski's theorem. Determine which of the graphs shown in figure are planar or non planar. [05]



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